Tie Zheng, Ph.D.

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Research focus: Air pollution exposure; Environmental health; Electrification and decarbonization; Air pollution modeling and measurements; Instrumentation and field campaigns

I combine multi-platform monitoring with data-driven models to quantify air pollution dynamics and human exposure in homes and cities. My goal is to provide strong scientific evidence to accelerate electrification and decarbonization efforts.

Outside research, I am a competitive distance runner and part-time coach. I placed 6th overall in the 2026 San Francisco Marathon (fastest Chinese runner).

Education

Tsinghua University — Ph.D. | 2021–2025 (conferred 06/19/2025) | Advisor: **Ye Wu** | GPA 3.82/4.0
Major in Environmental Science and Engineering; Minor in Big Data Project
Dissertation: *Fine-Grained Mapping of Air Pollution Using Mobile Monitoring and Machine Learning*

Shanghai Jiao Tong University — M.Eng., Transportation Engineering | 2021 | Advisor: **ZR Peng** | GPA 3.68/4.0
Dissertation: *Spatiotemporal Distributions of Traffic-related Pollutants in Urban Road Environments*

Chongqing University — B.Eng., Civil Engineering | 2018 | Advisor: **Zhitao Yan** | GPA 3.50/4.0

Research Experience

Stanford University — Postdoctoral Researcher, 08/2025–present | Advisor: **Rob Jackson** | Earth System Science
- ❖ Led Ultrafine Particle (UFP) project in Electrification for Health program: We report, to our knowledge, the first estimates of size-resolved UFP emissions from gas stoves across homes.
- ❖ Performed year-long UFP emission measurements in 60+ San Francisco Bay Area homes; derived energy-normalized, size-resolved UFP emission factors from residential gas appliances.

Tsinghua University — Doctoral Researcher, 2021–2025 | Advisor: **Ye Wu** | School of Environment
- ❖ Mapped hyperlocal air pollution in cities using machine learning and large-scale mobile measurement datasets (~1,500 h). Led multi-platform street-by-street mobile monitoring.
- ❖ Field campaign lead, NSFC-funded project: Hyperlocal air pollution field campaign in Macau.

Spatial Data Lab, Harvard University — Research Intern, 03/2025–09/2025 | Advisor: **Siqin Wang**
- ❖ Integrated large geospatial datasets with statistical and machine-learning methods for exposure mapping.

MIT — Visiting Researcher, 07/2024–12/2024 | Advisors: **Carlo Ratti, Simone Mora** | Urban Planning
- ❖ Data analyst for the City Scanner project: turning vehicles into sensing platforms.

Imperial College London — Visiting Researcher, 02/2024–06/2024 | Advisor: **Audrey de Nazelle**
- ❖ Worked on cost-effective mapping of hyperlocal air pollution, linking air pollution and human exposure.

Macau University of Science and Technology — Visiting Researcher, 01/2024–02/2024 | Advisor: **You Yan**
- ❖ Conducted field investigation and data collection for hyperlocal air quality in Macau, China.

Chinese Academy of Environmental Planning — Research Intern, 05/2022–08/2022 | Advisor: **Bofeng Cai**
- ❖ Worked on innovative technology for carbon peaking and carbon neutrality in China.

Publications

** Corresponding author; # equal contribution.*

Manuscripts under review or in preparation

[1] **Zheng, T.**, Derrick, S., Finnegan, C. J., Pongetti, J., Jourdain, C., Lebel, E. D., Kashtan, Y. S., Lucha, N., Huang, G., Michanowicz, D. R., Shonkoff, S. B. C., Boies, A. M., & Jackson, R. B. (2026). Ultrafine particle emissions from burning gas in homes. *Submission planned for October 2026*.

[2] Sumaiya P. K., **Zheng, T.***, Wen, Y., Xiang, S., et al. (2026). Multi-source domain adaptation for mobile-to-ambient air-pollution mapping. *Manuscript in preparation*.

Peer-reviewed articles and chapters

[1] **Zheng, T.**, Wen, Y., Xiang, S., et al. (2026). Reducing Sampling Effort for Mapping Air Pollution: A Two-Stage Framework. **Accepted.** *ACS ES&T Air.*

[2] **Zheng, T.**, Wen, Y., Xiang, S., Yang, P., Zheng, X., You, Y., Zhang, S., & Wu, Y. (2025). Cost-effective mapping of hyperlocal air pollution using large-scale mobile monitoring and land-use machine learning. *ACS ES&T Air*, 2(2), 151–161. **Journal cover.**

[3] Xiang, S., Yu, J., Yu, Y. T., et al., including **Zheng, T.** (2025). Exploring the holiday effect on elevated traffic-related air pollution with hyperlocal measurements in Chengdu, China. *Atmosphere*, 16(2), 171.

[4] Yu, Y. T., Zhang, S., Xiang, S., Yang, P., Wang, H., **Zheng, T.**, & Wu, Y. (2025). The role of ground-based mobile monitoring in air quality evaluation. In J. R. Barker, A. L. Steiner, & T. J. Wallington (Eds.), *Urban and regional air quality: Emissions, chemistry, and impacts* (pp. 267–323). World Scientific Publishing.

[5] Zhang, L., Cai, B., & **Zheng, T.** (2023). Developing an innovative technology system for carbon peaking and carbon neutrality. In J. Wang & H. Xu (Eds.), *Introduction to carbon dioxide peaking and carbon neutrality* (pp. 345–361). China Science and Technology Press. [In Chinese]

[6] **Zheng, T.**, Xiang, S., Zhang, S., & Wu, Y. (2023). Variability of traffic-related air pollutants at two- and four-phase intersections. *Atmospheric Pollution Research*, 14(12), 101936.

[7] **Zheng, T.**, Peng, Z. R., He, H. D., Zhang, S. J., & Wu, Y. (2022). Horizontal profiles of size-segregated particle number concentration and black carbon beside a major roadway. *Atmospheric Environment: X*, 16, 100187.

[8] **Zheng, T.#**, Jia, Y. P.#, Zhang, S., Li, X. B., Wu, Y., Wu, C. L., He, H. D., & Peng, Z. R. (2021). Impacts of vegetation on particle concentrations in roadside environments. *Environmental Pollution*, 282, 117067.

[9] **Zheng, T.**, Wang, H. W., Li, X. B., Peng, Z. R., & He, H. D. (2021). Impacts of traffic on roadside particle variations in varied temporal scales. *Atmospheric Environment*, 253, 118354.

[10] **Zheng, T.**, Li, B., Li, X. B., Wang, Z., Li, S. Y., & Peng, Z. R. (2021). Vertical and horizontal distributions of traffic-related pollutants beside an urban arterial road based on unmanned aerial vehicle observations. *Building and Environment*, 187, 107401.

[11] Jia, Y. P., Lu, K. F., **Zheng, T.**, Li, X. B., Liu, X., Peng, Z. R., & He, H. D. (2021). Effects of roadside green infrastructure on particle exposure: A focus on cyclists and pedestrians on pathways between urban roads and vegetative barriers. *Atmospheric Pollution Research*, 12(3), 1–12.

[12] Gao, Y., Wang, Z., Li, C., **Zheng, T.**, & Peng, Z. R. (2021). Assessing neighborhood variations in ozone and $PM_{2.5}$ concentrations using decision tree method. *Building and Environment*, 188, 107479.

[13] Wang, J., Yin, J., Khan, R. U., Wang, S., & **Zheng, T.** (2021). A study of inbound logistics mode based on JIT production in cruise ship construction. *Sustainability*, 13(3), 1588.

Presentations

- Stanford Global and Planetary Health Research Convening, 12th Annual (2026)
- International Symposium on Spatiotemporal Data Science (2026)
- Transportation Research Board, 104th Annual Meeting (2025)
- 7th International Students' Ecology and Environment Forum (2024) **Second Prize in Oral Presentation**
- UK and Ireland Exposure Science Meeting (2024)
- AGU24 Annual Meeting (2024)
- Urban Transitions (2024)
- 25th International Transport and Air Pollution Conference (2023)
- 16th and 17th National Environmental Conference for Doctoral Students, Tsinghua University (2022–2023)
- Transportation Research Board, 99th–102nd Annual Meetings (2020–2023)

Patent, Editorial Service, and Research Grants

- Co-inventor, "Air Quality Prediction System Based on Temporal Gate Mechanism," Chinese patent ZL 202310185638.6
- Guest Editor, *Atmosphere*, 2026–2027 Special Issue "Air Pollution Monitoring, Traffic-Related Emissions, and Health Exposure Assessment," Air Quality section; submissions open through March 2027
- Visiting Research Fund, Tsinghua University, 2024
- Conference Travel and Registration Support, TAP 2023, Gothenburg, Sweden, 2023
- Undergraduate Research Fund, *Wind-Induced Deflection of Suspension Insulators and Mitigation Technologies*, Chongqing University, 2017

Honors and Awards

- Three-time Tsinghua Comprehensive Scholarship, 2023–2025
- Tsinghua University Hash Scholarship, 2024
- Outstanding Team Award (Team leader), Big Data Project Practice Course, 2023
- Top 10 Athlete of the Year, Tsinghua University, 2023
- Outstanding Intern, Chinese Academy of Environmental Planning, 2022

Mentoring, Teaching and Leadership

- **Mentor**, Stanford SURGE Program, Doerr School of Sustainability, 2026
  Mentee: Sumaiya P. K., Gettysburg College
  Working Topic: *Mapping Air Health Inequity with AI and Community-Based Sensing*
- Stanford Postdoc Mentoring Certificate, 2026
- Selected Participant, Stanford Preparing Future Teaching Professors Program, 2026
- Running Coach, Palo Alto, CA, 2026–present
- Teaching Assistant, Tsinghua University, 2023–2025
- Team leader, Big Data Project Practice Course, Tsinghua University, 2023
- Vice-Captain, Tsinghua University Running Team, 2023

Peer Review

- **Reviewer for 23 academic journals**, 2021–present
- **Atmospheric science and air quality (4):** Air; Atmosphere; Atmospheric Environment; Atmospheric Environment: X
- **Environmental science, sustainability, and public health (7):** International Journal of Phytoremediation; Journal of Hazardous Materials; Pollutants; Science of the Total Environment; Sustainability; Toxics; Trends in Public Health
- **Buildings, cities, and land use (4):** Building and Environment; Land; Smart Cities; Urban Science
- **Transportation and energy (3):** Journal of Traffic and Transportation Engineering (English Edition); Transportation Research Part D: Transport and Environment; World Electric Vehicle Journal
- **Computing, engineering, and multidisciplinary science (5):** Algorithms; Applied Sciences; Automation; Earth Science Informatics; Scientific Reports

Technical Skills

- **Measurement and Data Collection:** Multi-pollutant high-resolution monitoring; mobile and UAV-based monitoring; instrumentation and platform development
- **Exposure and Modeling:** Exposure characterization; inhaled-intake estimation; multizone modeling; land-use regression; machine learning; spatiotemporal statistics; uncertainty and sensitivity analysis
- **Computing:** Python (advanced), R, SQL, ArcGIS Pro, LaTeX
- **Languages:** Mandarin (native) and English (fluent)

Athletic Achievements

Personal bests: 10K 30:14; half marathon 1:08:37; marathon 2:27:01

- 6th overall, San Francisco Marathon (Full), 2026
- 5th, San Jose Half Marathon, 2026
- 3rd, Rock ‘n’ Roll San Jose 10K (road race), 2025
- 1st (**meet record**), Men’s 10,000m, Beijing Collegiate Track and Field Championships, 2022, 2024
- 1st (**meet record**), Men’s 5,000m, London University Outdoor Championships, 2024
- 1st (**meet record**), Men’s marathon, Huai Rou Great Wall Marathon, 2023
- 1st overall, Victoria Park Half Marathon, London, 2024
- 1st, Men’s 5K (road race), Boston Mile for Migraine Race, 2024
- 1st, Men’s 100m Backstroke, Tsinghua University Swimming Competition, 2022
- 1st (**meet record**), Men’s 800m, 1,500m, 5,000m, Tsinghua University Track and Field Championships, 2022, 2023, 2024